

NEWTON SCHOOL OF TECHNOLOGY

Data Visualization & Analytics — Capstone 2

Brazilian E-Commerce Analytics

Analysing Order Fulfilment, Delivery Performance,
Customer Satisfaction & Revenue Growth
for a Large Online Marketplace (2016–2018)

Sector	E-Commerce / Retail
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1. Executive Summary

This report presents the findings of the DVA Capstone 2 project conducted by Section B, Group G-7 at Newton School of Technology. The project analyses the Olist Brazilian E-Commerce dataset, which spans over 114,000 orders placed between 2016 and 2018 across Brazil's largest multi-seller online marketplace.

Business Challenge: A large online marketplace needed to identify the key operational drivers of delivery delays, poor customer reviews, order cancellations, and revenue leakage — and determine where to intervene first to improve service quality and customer retention.

Approach: The team sourced raw transactional data from Kaggle, executed a complete Python-based ETL pipeline, performed exploratory and statistical analysis in Jupyter Notebooks, and built a three-page interactive Tableau Public dashboard covering Sales, Geography, and Delivery operations.

Key Findings:

- Total platform revenue reached **\$18.1 million** across 103,430 orders, growing steadily from 2016 to 2018.
- **7.2% of all orders** (8,756) were delivered late, with an average delay of approximately 9 days beyond the estimated date.
- The **on-time delivery rate** was 92.3%, though certain categories and seller states significantly underperformed this benchmark.
- **Credit card** dominated payment, accounting for 76.8% of revenue (\$13.9M), with an average of 3 instalments per transaction.
- **São Paulo (SP)** was the dominant state for both sellers and customers, generating the highest revenue and order volume by a wide margin.
- Average customer review score was **4.03 out of 5.0**, with a clear inverse relationship between delivery time and satisfaction.

Key Recommendations:

1. Implement targeted logistics interventions for high-delay seller states (SP, PR, MG) and slow-delivery categories.
2. Establish seller performance scorecards to monitor and remediate chronic underperformers.

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3. Introduce proactive communication and estimated delivery accuracy improvements to protect review scores.

2. Sector & Business Context

2.1 Sector Overview

Brazilian e-commerce has experienced exceptional growth, driven by increasing smartphone penetration, expanding logistics networks, and a growing middle-class consumer base. Olist is one of Brazil's largest online retail platforms, operating as a multi-vendor marketplace that connects small and medium businesses to major e-commerce channels. The platform manages the full order lifecycle from placement through payment, fulfilment, and delivery — making it a rich source of operational data.

2.2 Industry Challenges

The Brazilian e-commerce sector faces several structural challenges that this analysis directly addresses:

- **Logistics fragmentation:** Brazil's vast geography creates significant variation in delivery times across states, with remote regions often facing extended lead times.
- **Seller quality inconsistency:** Multi-vendor platforms depend on diverse seller networks, creating wide variance in fulfilment quality and customer experience.
- **Customer trust and retention:** Repeat purchase rates are sensitive to first-experience quality, particularly delivery reliability and post-purchase support.
- **Payment complexity:** The prevalence of instalment-based credit card payments (boleto and instalment credit) introduces payment timing risk and potential revenue realisation delays.

2.3 Decision-Maker Profile

This project serves the Chief Operating Officer (COO) and Head of Seller Management of a large Brazilian online marketplace. The findings are designed to inform decisions on logistics prioritisation, seller onboarding standards, customer retention investment, and category-level fulfilment strategy.

2.4 Business Value

Improving delivery reliability by even a few percentage points can meaningfully increase repeat purchase rates, reduce complaint volumes, and strengthen the platform's seller reputation — directly translating to higher lifetime customer value and reduced operational cost from returns and refunds.

3. Problem Statement & Objectives

3.1 Formal Problem Definition

A large online marketplace wants to improve delivery reliability, customer satisfaction, and seller performance by analysing order status, payment behaviour, product attributes, delivery timelines, and customer review scores. The company needs to identify the main operational factors that cause delays, complaints, and revenue loss so it can make better decisions about fulfilment, seller management, and customer experience.

Core Business Question:

Which products, sellers, regions, and delivery patterns should management focus on to reduce service issues and improve revenue and repeat business?

3.2 Project Scope

In scope:

- Order-level analysis covering status, payment, delivery, and review dimensions
- Seller and category-level performance benchmarking
- Geographic segmentation by Brazilian state
- KPI computation and statistical analysis of delivery and satisfaction relationships
- Tableau dashboard design for executive and operational audiences

Out of scope:

- Individual customer profiling or PII-based analysis
- Real-time monitoring or predictive modelling
- Product-image or text-based recommendation systems

3.3 Success Criteria

The project is considered successful if:

1. Five KPIs are computed and validated from the cleaned dataset.
2. At least one statistically supported relationship between delivery and satisfaction is identified.
3. A three-view interactive Tableau dashboard is published and accessible via a public URL.
4. Three to five specific, actionable business recommendations are delivered with estimated

impact.

4. Data Description

4.1 Dataset Source

Attribute	Details
Source	Kaggle — Olist Brazilian E-Commerce Public Dataset
Direct Link	kaggle.com/datasets/olistbr/brazilian-ecommerce
Total Orders	≈114,092 orders
Row Count	100,000+ rows (primary orders table)
Merged Column Count	37 columns after joining key tables
Time Period	September 2016 – August 2018
Format	Multiple CSVs (relational schema)

4.2 Data Structure

The Olist dataset follows a normalised relational schema consisting of nine interlinked CSV files. The primary tables merged for this analysis were:

`orders`, `order_items`, `order_payments`, `order_reviews`, `customers`, `sellers`, `products`, and `geolocation`.

4.3 Key Columns Used

Column Name	Description	Role in Analysis
order_id	Unique order identifier	Join key across tables
customer_id	Unique customer per order	Customer-level analysis
order_purchase_timestamp	Order placement datetime	Time trends & monthly KPIs
order_delivered_customer_date	Actual delivery datetime	Delay computation
order_estimated_delivery_date	Estimated delivery date shown to customer	Delivery gap analysis
order_status	Current status of the order	Cancellation & fulfilment
review_score	1–5 customer satisfaction rating	Customer experience KPI
payment_value	Total payment per order	Revenue analysis
payment_type	Method of payment	Payment behaviour
product_category_name	Product category	Category performance
seller_id	Unique seller identifier	Seller performance
customer_state	Customer state (Brazilian UF code)	Geographic segmentation
seller_state	Seller state (Brazilian UF code)	Regional fulfilment

4.4 Data Limitations

- **Missing review data:** A subset of delivered orders have no associated review score, limiting satisfaction analysis to reviewed orders only.
- **Geolocation approximation:** The geolocation table provides ZIP-code-level coordinates, which may introduce minor inaccuracies in geographic aggregation.
- **Time truncation:** Data ends in August 2018, meaning seasonal patterns for late 2018 cannot be observed.
- **Single marketplace:** Findings reflect one platform's operations and may not generalise to all Brazilian e-commerce contexts.

5. Data Cleaning & ETL Pipeline

All data cleaning and transformation steps were executed in Python using Jupyter Notebooks, as required by the Capstone 2 specification. The pipeline is documented and committed to the GitHub repository under `notebooks/02_cleaning.ipynb`.

5.1 Step 1: Data Loading & Initial Inspection

The nine Olist CSV files were loaded using `pandas.read_csv()` and inspected for shape, data types, null counts, and duplicate records. An initial schema diagram was validated to confirm correct join keys (`order_id`, `customer_id`, `seller_id`, `product_id`).

5.2 Step 2: Handling Missing Values

Column	Missing %	Treatment
<code>order_delivered_customer_date</code>	~3.1%	Retained as NaN; excluded from delivery time calculations
<code>review_score</code>	~0.8%	Excluded from review aggregations; flagged as Null in dashboard
<code>product_category_name</code>	~0.1%	Labelled “Unknown” for category grouping
<code>order_approved_at</code>	~0.2%	Imputed with median approval delay per order status

5.3 Step 3: Outlier Detection & Treatment

- **Payment values:** Extreme high-value transactions (above the 99th percentile, > \$5,000) were investigated and retained as legitimate bulk orders.
- **Delivery times:** Negative delivery durations (actual delivery before purchase) were flagged as data quality errors and excluded from delivery analysis.
- **Review scores:** Scores outside the 1–5 range were absent; the distribution was validated clean.

5.4 Step 4: Data Type Conversions

All timestamp columns (`order_purchase_timestamp`, `order_delivered_customer_date`, `order_estimated_delivery_date`) were parsed to `datetime64` using `pd.to_datetime()`. Brazilian state codes were standardised to uppercase two-letter UF codes.

5.5 Step 5: Feature Engineering

The following derived columns were created to support KPI computation and Tableau visualisation:

Derived Column	Logic
<code>delivery_days</code>	<code>order_delivered_customer_date</code> – <code>order_purchase_timestamp</code> (in days)
<code>delivery_delay_days</code>	<code>order_delivered_customer_date</code> – <code>order_estimated_delivery_date</code> (positive = late)
<code>late_flag</code>	Binary: 1 if <code>delivery_delay_days</code> > 0, else 0
<code>order_month</code>	Month extracted from <code>order_purchase_timestamp</code>
<code>order_year</code>	Year extracted from <code>order_purchase_timestamp</code>
<code>order_quarter</code>	Quarter extracted from <code>order_purchase_timestamp</code>
<code>revenue_clean</code>	Aggregated payment values per order after null-payment exclusion

5.6 Step 6: Table Merging

The cleaned individual tables were merged into a single analysis-ready master dataframe using `pd.merge()` on appropriate keys, producing a 37-column dataset with approximately 100,000+ order-item-level records. The cleaned output was saved to `data/processed/olist_cleaned_master.csv`.

6. KPI & Metric Framework

Five KPIs were selected to track the core dimensions of the business problem. Each was computed in Python (`notebooks/05_final_load_prep.ipynb`) and surfaced in the Tableau dashboard.

KPI	Definition	Formula
Monthly Revenue Growth %	Revenue change month-over-month	$\frac{\text{Current MoM Revenue} - \text{Previous MoM Revenue}}{\text{Previous MoM Revenue}} \times 100$
Avg Customer Review Score	Customer satisfaction level	$\overline{\text{review_score}}$ across all reviewed orders
Late Delivery Rate	Share of orders delivered past estimate	$\frac{\text{Orders with late_flag}=1}{\text{Total delivered orders}} \times 100$
Cancellation Rate	Share of orders not fulfilled	$\frac{\text{Cancelled orders}}{\text{Total orders}} \times 100$
On-Time Delivery Rate	Share of orders delivered by or before estimated date	$\frac{\text{On-time delivered orders}}{\text{Total delivered orders}} \times 100$

Observed KPI Values (2016–2018, all data):

Total Revenue	Total Orders	Avg Order Value	Avg Review Score	On-Time Rate
\$18.1M	103,430	\$175	4.03 / 5.0	92.3%

7. Exploratory Data Analysis

7.1 Revenue Trends

Monthly cumulative revenue grew steadily from early 2017 through mid-2018, reaching a peak of approximately \$10M (running cumulative) by August 2018. The Year-on-Year quarterly comparison reveals:

- Q1 2018 generated \$3.74M vs. Q1 2017's \$0.94M — a nearly 4× increase.
- Q2 remained the highest-revenue quarter in both 2017 (\$3.84M) and 2018 (comparable levels).
- Q4 2016 showed minimal revenue (\$0.06M), reflecting the platform's early growth stage.

Business Insight: Revenue growth is accelerating strongly year-over-year. 2018's quarterly figures suggest a platform reaching maturity, but Q3 and Q4 2018 show signs of levelling, warranting investigation into whether this reflects seasonal patterns or growth deceleration.

7.2 Order Status Distribution

The vast majority of orders (more than 97%) were in a delivered status. Cancelled orders represent a small but non-trivial share. Processing, shipped, and unavailable statuses together form a minor residual. This distribution confirms that the platform's fulfilment pipeline is broadly functional, but the quality of delivered orders (as measured by review score and delivery timeliness) is the primary lever for improvement.

7.3 Top Revenue Categories

Product Category	Revenue	Rank
bed_bath_table	\$1.64M	1
computers_accessories	\$1.41M	2
health_beauty	\$1.37M	3
furniture_decor	\$1.30M	4
sports_leisure	\$1.24M	5
watches_gifts	\$1.22M	6
housewares	\$1.01M	7
garden_tools	\$0.74M	8
auto	\$0.72M	9
cool_stuff	\$0.68M	10

Business Insight: The top three categories (bed/bath/table, computers/accessories, health/beauty) collectively account for a disproportionate share of revenue. These categories warrant priority fulfilment investment and seller quality oversight.

7.4 Payment Behaviour

Credit card is the dominant payment method, contributing \$13.9M (76.8%) of total revenue. Boleto (bank slip) follows at \$3.6M (19.9%). Voucher and debit card together account for less than 4%. The average number of credit card instalments per transaction is 3.0×, indicating that Brazilian consumers heavily utilise instalment-based purchasing — a structural feature of the local market.

7.5 Geographic Distribution

- **Top customer state:** São Paulo (SP) with 48,093 orders, followed by Minas Gerais (MG) at 13,301 and Rio de Janeiro (RJ) at 14,758.
- **Top seller state by revenue:** SP dominates at \$11.87M in seller-side revenue, followed by PR (\$1.65M) and MG (\$1.38M).
- **Customer-side revenue:** SP leads at \$7.55M, followed by RJ (\$2.71M) and MG (\$2.27M).

The geographic concentration in the South-East is consistent with Brazil's economic geography, where SP and RJ represent the largest consumer and logistics hubs.

7.6 Review Score Distribution

Of orders with review data, 58,084 received a 5-star rating and 19,267 received a 4-star rating — together representing over 75% of reviewed orders. However, 13,194 orders received a 1-star rating, signalling a significant dissatisfied customer segment that merits targeted investigation.

8. Statistical Analysis

8.1 Delivery Time vs Review Score

The scatter plot of average delivery days against average review score (aggregated at the monthly level) reveals a clear negative relationship: as average delivery time increases, average review score decreases. The trend lines segment orders into three delivery speed groups:

- **Fast delivery (<8 days):** Higher average review scores, clustering above 4.0.
- **Medium delivery (8–13 days):** Mixed review scores, average around 3.8–4.1.
- **Slow delivery (>13 days):** Notably lower review scores, with some data points below 3.0, particularly for the slowest deliveries.

Statistical Finding: The negative slope of the trend line in the Slow delivery segment is steeper than in the Fast segment, suggesting a non-linear relationship: delivery delays have a disproportionately large negative impact on customer satisfaction beyond a threshold of approximately 13 days.

8.2 Late Order Volume by Month

Monthly delivery performance analysis reveals that late order counts are not uniformly distributed across the year. Certain months — notably January, February, March, and November — show elevated late order counts (1,260; 554; 1,812; 267 respectively), while mid-year months (June, July, August) show lower late counts, possibly reflecting reduced order volumes during those periods.

Average delivery days remain relatively stable across months (ranging from approximately 9.4 to 14.8 days), with a slight upward trend toward year-end, consistent with holiday demand surges.

8.3 Late Orders by Seller State

São Paulo (SP) accounts for 6,730 late orders — disproportionately high relative to other states. However, this partly reflects SP's dominant order volume. Minas Gerais (MG) at 481 and Paraná (PR) at 550 also show notable late order counts. States such as Espírito Santo (ES, 24) and Bahia (BA, 34) have low absolute late order counts.

Key Finding: SP's late order count is over 12× that of the next highest state (RJ at 382). While partially explained by volume, SP sellers' delivery rate warrants targeted quality review given the state also dominates revenue generation.

8.4 Delivery Performance by Category

Average delivery days vary significantly by product category. Categories performing above the overall average (≈ 12 days) include audio, agro-industry items, and art categories — potentially due to heavier items requiring specialist logistics. Categories performing below average include arts-and-craftsmanship items, which likely ship as lightweight parcels.

9. Tableau Dashboard Design

9.1 Dashboard Objective

The Tableau dashboard was designed to answer the core business question at two levels: an executive summary for revenue and satisfaction KPIs, and an operational drill-down for geographic, seller, and delivery performance. The dashboard was published on Tableau Public and is accessible via a public URL.

9.2 Dashboard Structure

The dashboard comprises three pages, each with interactive filters for Year, State, Status, Payment Type, Category, and Quarter.

9.2.1 Page 1: Sales Overview



Figure 9.1. Sales Overview — Revenue trends, order status, payment type, and top categories

KPI Cards: Total Revenue (\$18.1M), Total Orders (103.43K), Average Order Value (\$175), Average Review Score (4.03).

Charts included:

- Monthly Revenue Trend (cumulative running sum, 2016–2018)
- Order by Status (donut chart)
- Revenue by Payment Type (bar chart)
- Top Categories Revenue (horizontal bar chart)
- Quarterly Revenue Year-on-Year Comparison (grouped bar chart)

9.2.2 Page 2: Customers & Geography



Figure 9.2. Customers & Geo — Geographic order distribution, review scores, and payment breakdown

KPI Cards: Unique Customers (86.88K), Top State (SP), Credit Card % (75.8%), Avg Instalments (3.0×).

Charts included:

- Orders by Customer State (choropleth map of Brazil)
- Top 10 States by Order Volume (bar chart)
- Payment Type Distribution (donut chart)
- Seller State Revenue (bar chart)
- Customer State Revenue (bar chart)
- Review Score Distribution (bar chart)

9.2.3 Page 3: Delivery & Operations



Figure 9.3. Delivery & Ops — On-time rate, late orders by state and category, and review-delivery relationship

KPI Cards: On-Time Rate (92.3%), Avg Delivery Time (12.0 days), Late Orders (8,756 / 7.2% of total), Avg Delay when Late (~9 days).

Charts included:

- On-Time vs Late Deliveries (donut chart)
- Avg Delivery Days by Category (colour-coded bar: above / near / below average)
- Late Orders by Seller State (bar chart)
- Review Score vs Delivery Time (scatter plot with trend lines by speed segment)
- Monthly Delivery Performance — Dual Axis (bar + line: late count + avg delivery days)

9.3 Interactive Elements

Six global filters are present across all dashboard pages: Year, State, Order Status, Payment Type, Product Category, and Quarter. The dashboard does not hard-code any numbers; all KPI cards and charts update dynamically based on filter selections.

10. Insights Summary

The following insights are drawn from the combined EDA, statistical analysis, and dashboard exploration. Each insight is stated in decision language.

1. **Revenue is growing strongly but concentration is high.** Total revenue reached \$18.1M, with Q1 2018 alone generating nearly 4× the revenue of Q1 2017. However, the top three product categories (bed/bath/table, computers/accessories, health/beauty) drive a disproportionate share of revenue. Disruption in any of these categories — through stockouts, quality issues, or logistics failures — could materially impact platform revenue.
2. **Credit card instalments dominate and create payment timing risk.** With 75.8% of revenue flowing through credit card and an average of 3.0 instalments per transaction, the platform faces a significant payment realisation lag. Revenue shown at order time may take months to fully settle, affecting cash-flow planning for sellers.
3. **7.2% of all orders are late, with an average overrun of ~9 days.** While the 92.3% on-time rate appears strong, the 8,756 late orders represent meaningful service failures. A 9-day average overrun — relative to an already-estimated 12-day average delivery time — means some customers wait more than three weeks from order placement, a figure likely to drive churn.
4. **Late deliveries directly and disproportionately reduce customer satisfaction.** The scatter analysis confirms that as average delivery time exceeds 13 days, review scores decline more steeply. Orders in the “slow” delivery band receive visibly lower scores than those in the “medium” band, and the relationship is non-linear: each additional day of delay has greater marginal impact at higher delay levels.
5. **São Paulo sellers account for 76.8% of all late orders.** SP-based sellers generate 6,730 of the 8,756 total late orders. While SP dominates order volume, this concentration suggests that logistics routing and seller compliance within SP specifically are the highest-priority targets for operational intervention.
6. **The review score distribution is bimodal, skewed toward satisfaction.** 56.4% of orders receive a 5-star review and 18.7% receive a 4-star review. However, 12.8% receive a 1-star review — indicating a significant dissatisfied tail. The gap between 1-star and 2-star volume (13,194 vs. 3,484) suggests that the most dissatisfied customers tend to report extreme disappointment rather than moderate dissatisfaction, consistent with delivery-related failures.
7. **São Paulo is the dominant revenue state on both the seller and customer side.** SP

sellers generate \$11.87M in revenue (seller-side) and SP customers spend \$7.55M (customer-side). The state's dominance means any logistics or policy decision applied uniformly across Brazil will disproportionately affect SP outcomes — both positively and negatively.

8. **Q2 is consistently the highest-revenue quarter.** Both 2017 and 2018 show Q2 as the peak revenue quarter (\$3.84M in 2017, comparable in 2018). Operations, seller readiness, and customer service capacity should be scaled in advance of Q2 each year.
9. **Certain product categories exhibit above-average delivery times.** Audio, agro-industry, and art categories show average delivery times materially above the 12-day platform average, likely due to product weight, specialist handling, or remote seller locations. These categories may benefit from dedicated logistics agreements or adjusted estimated delivery dates.
10. **Boleto (bank slip) payments represent a meaningful secondary channel at 19.9%.** Boleto orders may carry higher cancellation risk if customers initiate but do not complete payment. Monitoring boleto conversion rates and comparing cancellation rates to credit card orders would refine revenue forecasting.

11. Business Recommendations

Recommendation 1: Establish a Seller Delivery Compliance Programme for SP-Based Sellers

Insight: São Paulo sellers generate 76.8% of all late orders despite also being the platform's highest revenue contributors.

Action: Implement a tiered seller scorecard system that tracks on-time delivery rate, average delay days, and review score per seller on a monthly basis. Sellers falling below a 90% on-time threshold trigger an automatic compliance review. Chronic underperformers (below 85% for two consecutive months) face temporary listing restrictions and mandatory logistics consultation.

Expected Impact: A 10% reduction in SP late orders would eliminate ≈ 673 service failures per period, with estimated review score improvement of 0.1–0.2 points on the overall platform average — directly improving customer retention probability.

Recommendation 2: Improve Delivery Estimation Accuracy and Proactive Communication

Insight: The average delay for late orders is ~ 9 days beyond the estimated date. Poor estimate accuracy is likely a key driver of customer disappointment, as customers form expectations based on the displayed estimate.

Action: Audit the delivery estimation algorithm by seller state and product category. Adjust estimated delivery dates to incorporate historical average actual delivery times for each state-category combination. Implement automated customer notifications triggered at 80% of the estimated delivery window if the order has not yet shipped.

Expected Impact: Resetting customer expectations through more accurate estimates can reduce 1-star reviews driven by “later than expected” complaints by an estimated 15–20%, without requiring any change in actual logistics operations.

Recommendation 3: Prioritise Fulfilment Investment in Top-3 Revenue Categories

Insight: Bed/bath/table, computers/accessories, and health/beauty together drive the highest revenue but are also among the categories with higher delivery day averages.

Action: Negotiate dedicated logistics SLAs with carriers for these three categories. Incentivise sellers in these categories to pre-position inventory in regional fulfilment centres.

Monitor category-level late rate separately from the platform average.

Expected Impact: Improving on-time delivery in the top-3 categories by 5 percentage points would protect an estimated \$200K–\$400K in revenue at risk from dissatisfied customers reducing repeat orders.

Recommendation 4: Build a Retention Campaign for Dissatisfied (1-Star) Customers

Insight: 13,194 orders received a 1-star review — a significant dissatisfied customer cohort with high churn probability.

Action: Identify 1-star reviewers and deploy a targeted recovery campaign within 48 hours of review submission: personalised apology message, discount voucher for next purchase, and expedited resolution of any reported complaint. Track re-purchase rate within 90 days for the intervention group vs. a control group.

Expected Impact: If 15% of 1-star reviewers can be recovered to make a second purchase at the average order value of \$175, the incremental revenue opportunity is approximately \$346K over the analysis period.

Recommendation 5: Investigate Q2 Demand Surge and Scale Operations Proactively

Insight: Q2 is consistently the highest-revenue quarter in both 2017 and 2018, yet late order counts remain elevated in February and March (adjacent quarters).

Action: Conduct a root-cause analysis of Q1/Q2 late order spikes. Implement seller inventory readiness checks 6 weeks before Q2 each year. Pre-negotiate additional carrier capacity for the January–March period to absorb post-holiday and pre-mid-year demand.

Expected Impact: Reducing Q1 late orders by 20% would improve on-time rate to approximately 93.7% and prevent ≈ 350 avoidable service failures per quarter.

12. Impact Estimation

#	Recommendation	Estimated Benefit Type	Estimated Magnitude
1	SP Seller Compliance Programme	Delivery quality improvement	−673 late orders/period; +0.1–0.2 review score
2	Estimation & Communication Fix	Customer satisfaction / churn reduction	15–20% reduction in expectation-driven 1-star reviews
3	Top-3 Category SLA Investment	Revenue protection	\$200K–\$400K at-risk revenue protected per year
4	1-Star Customer Recovery Campaign	Revenue recovery	~\$346K incremental revenue (15% recovery rate)
5	Q2 Demand Scaling	Late order reduction	≈350 fewer late orders per Q1–Q2 period

Note: All estimates are approximate and based on observed data trends. Actual outcomes will depend on implementation quality, seller cooperation, and external logistics factors. The logic underlying each estimate is documented in the Insights and Recommendations sections above.

13. Limitations

- **No causality established:** The observed relationship between delivery time and review score is correlational. Other factors (product quality, seller communication, packaging) also influence review scores and are not controlled for in this analysis.
- **Customer repeat purchase data:** The dataset provides customer IDs per order but does not directly link customers across orders without additional deduplication, limiting robust repeat purchase rate computation.
- **Seller anonymisation:** Seller IDs are anonymised, preventing qualitative assessment of individual seller characteristics or external reputation data.
- **Data ends August 2018:** Full-year 2018 analysis is not possible. Q3 and Q4 2018 trends, including any holiday-season effects, cannot be observed.
- **Product-level granularity:** Revenue and delivery analysis is at the order level; product-level unit economics (margin, return rate) are not available in the dataset.

14. Future Scope

- **Predictive late-delivery model:** A classification model (logistic regression or gradient boosting) trained on seller state, product category, payment type, and order timing could predict late delivery risk at the point of order placement — enabling proactive intervention before the delivery window closes.
- **Seller segmentation:** K-means clustering on seller-level KPIs (on-time rate, average review score, revenue, order volume) would identify distinct seller performance tiers and enable differentiated support programmes.
- **Review text sentiment analysis:** The dataset includes written review comments. NLP-based sentiment analysis could provide richer insight into the specific complaints driving 1-star ratings.
- **Real-time dashboard:** With an API-connected data pipeline, the Tableau dashboard could be extended to near-real-time operational monitoring, alerting operations teams to emerging late-delivery spikes before they scale.
- **Customer lifetime value modelling:** Linking repeat purchase behaviour to first-delivery experience would allow the business to quantify the long-term cost of each late delivery in terms of lost customer lifetime value — strengthening the business case for logistics investment.

15. Conclusion

This project delivered a full-cycle analytics engagement on the Olist Brazilian E-Commerce dataset, covering data sourcing, Python-based ETL, exploratory and statistical analysis, and an interactive three-page Tableau dashboard. The analysis revealed that while the platform has achieved strong revenue growth — reaching \$18.1M across 114,092 orders — operational reliability remains the primary lever for improving customer satisfaction and long-term retention. Late deliveries, concentrated in São Paulo-based sellers and slow-delivery product categories, directly suppress review scores and increase churn risk. Five prioritised recommendations, centred on seller compliance, communication accuracy, category-level SLA investment, customer recovery, and seasonal scaling, collectively address the most material causes of service failure. Implementing these actions is estimated to protect or recover several hundred thousand dollars in annual revenue while meaningfully improving the platform’s average review score and customer retention rate.

A. Data Dictionary

Column Name	Data Type	Description
order_id	string (UUID)	Unique identifier for each order
customer_id	string (UUID)	Unique customer per order (not a persistent customer ID)
customer_unique_id	string (UUID)	Persistent customer identifier across orders
order_status	categorical	Status: delivered, shipped, cancelled, etc.
order_purchase_timestamp	datetime	Date and time order was placed
order_approved_at	datetime	Date and time payment was approved
order_delivered_carrier_date	datetime	Date order was handed to carrier
order_delivered_customer_date	datetime	Date order was delivered to customer
order_estimated_delivery_date	datetime	Estimated delivery date shown to customer
payment_sequential	integer	Payment instalment number
payment_type	categorical	Payment method: credit_card, boleto, voucher, debit_card
payment_installments	integer	Number of payment instalments
payment_value	float	Total payment amount for the order
review_id	string (UUID)	Unique identifier for each review
review_score	integer (1–5)	Customer satisfaction score
review_comment_title	string	Short review title (often empty)
review_comment_message	string	Full review text comment
product_id	string (UUID)	Unique product identifier
product_category_name	categorical	Portuguese product category name
product_weight_g	float	Product weight in grams

Column Name	Data Type	Description
product_length_cm	float	Product length in centimetres
product_height_cm	float	Product height in centimetres
product_width_cm	float	Product width in centimetres
seller_id	string (UUID)	Unique seller identifier (anonymised)
seller_city	string	City of the seller
seller_state	string (UF code)	Brazilian state of the seller
customer_city	string	City of the customer
customer_state	string (UF code)	Brazilian state of the customer
delivery_days	float (derived)	Days from order placement to delivery
delivery_delay_days	float (derived)	Days late relative to estimated date (positive = late)
late_flag	binary (derived)	1 if order was delivered late; 0 otherwise
order_month	integer (derived)	Month of order placement
order_year	integer (derived)	Year of order placement
order_quarter	integer (derived)	Quarter of order placement
revenue_clean	float (derived)	Cleaned and aggregated payment value per order

B. Contribution Matrix

Team Member	Dataset & Sourc- ing	ETL & Clean- ing	EDA & Anal- ysis	Statistic Anal- ysis	Tableau Dash- board	Report Writ- ing	PPT
Saumya Kumar	Owner	Support	Support	None	Support	Support	Support
Saman Iqbal Ansari	Support	Owner	Owner	Support	Owner	None	None
Arina Ali	Support	Owner	None	None	Support	Owner	None
Yash Kumar	None	None	None	Owner	Owner	None	None
Ayush Sahu	None	None	None	None	None	None	Owner
Prakhar Srivas- tava	None	None	Owner	None	None	None	None

Declaration: We confirm that the above contribution details are accurate and verifiable through GitHub Insights, PR history, and submitted artefacts.

Team Lead: Saumya Kumar

Date: April 2026

Mentor: Satyaki Das

Institute: Newton School of Technology

Section: B **Group:** G-7

GitHub: github.com/oksaumya/SectionB_G7_BrazilianE-Commerce